Unified Modeling Language (UML) Structural & Behavioral Models

Academic Grievance, No-Dues & Clearance Automation Engine

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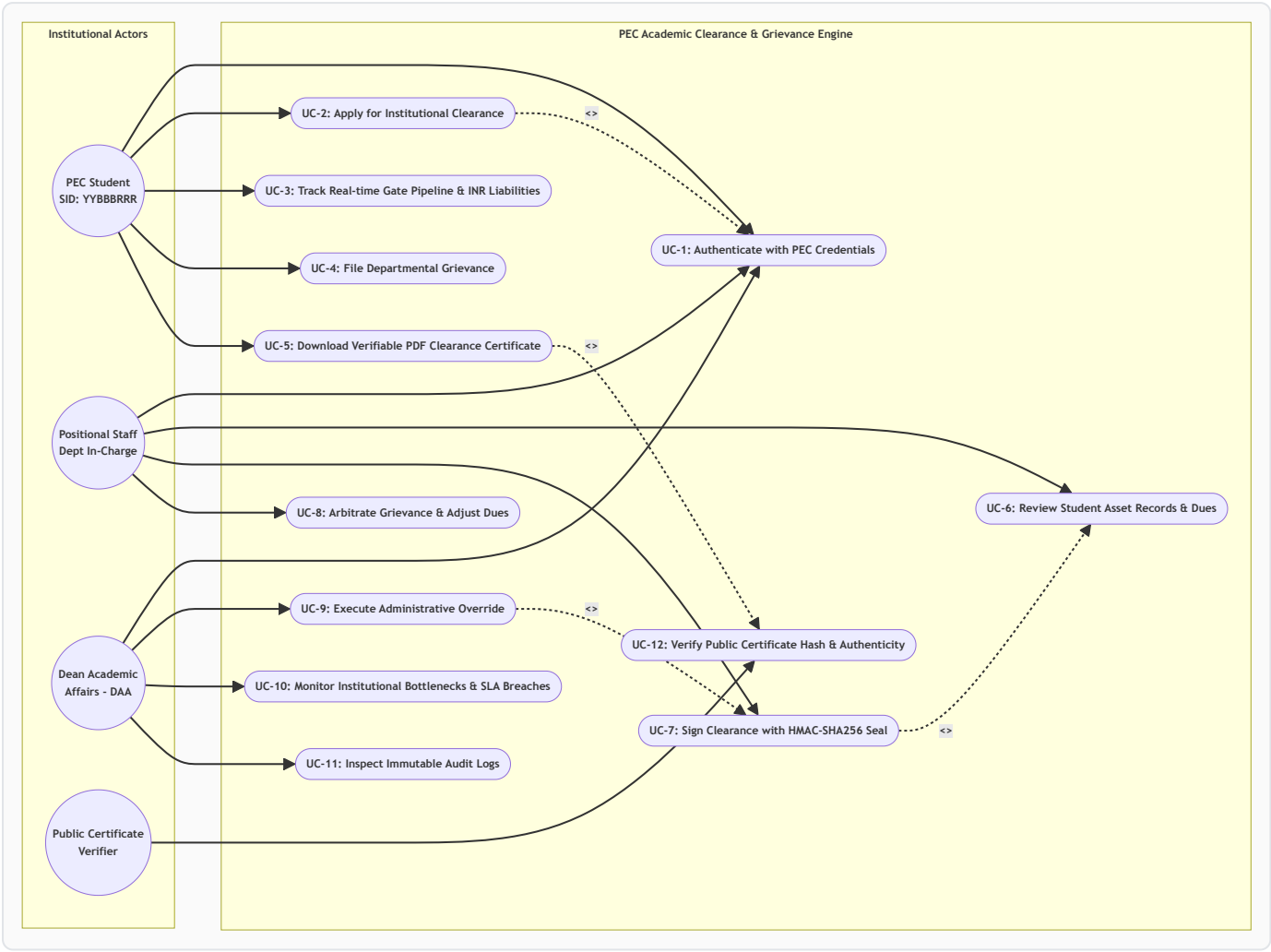
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1. Use Case Model & Diagram

1.1 Actors & Role Boundaries

- **Student Actor:** Registered PEC student identified by an 8-digit SID (e.g., `22103042`).
- **Positional Staff Actor:** Departmental authority operating under institutional designations.
- **Dean Academic Affairs (DAA / Admin Actor):** Executive authority holding override privileges and institutional audit oversight.
- **Public Certificate Verifier Actor:** External unauthenticated party validating clearance credentials.

1.2 Use Case Diagram

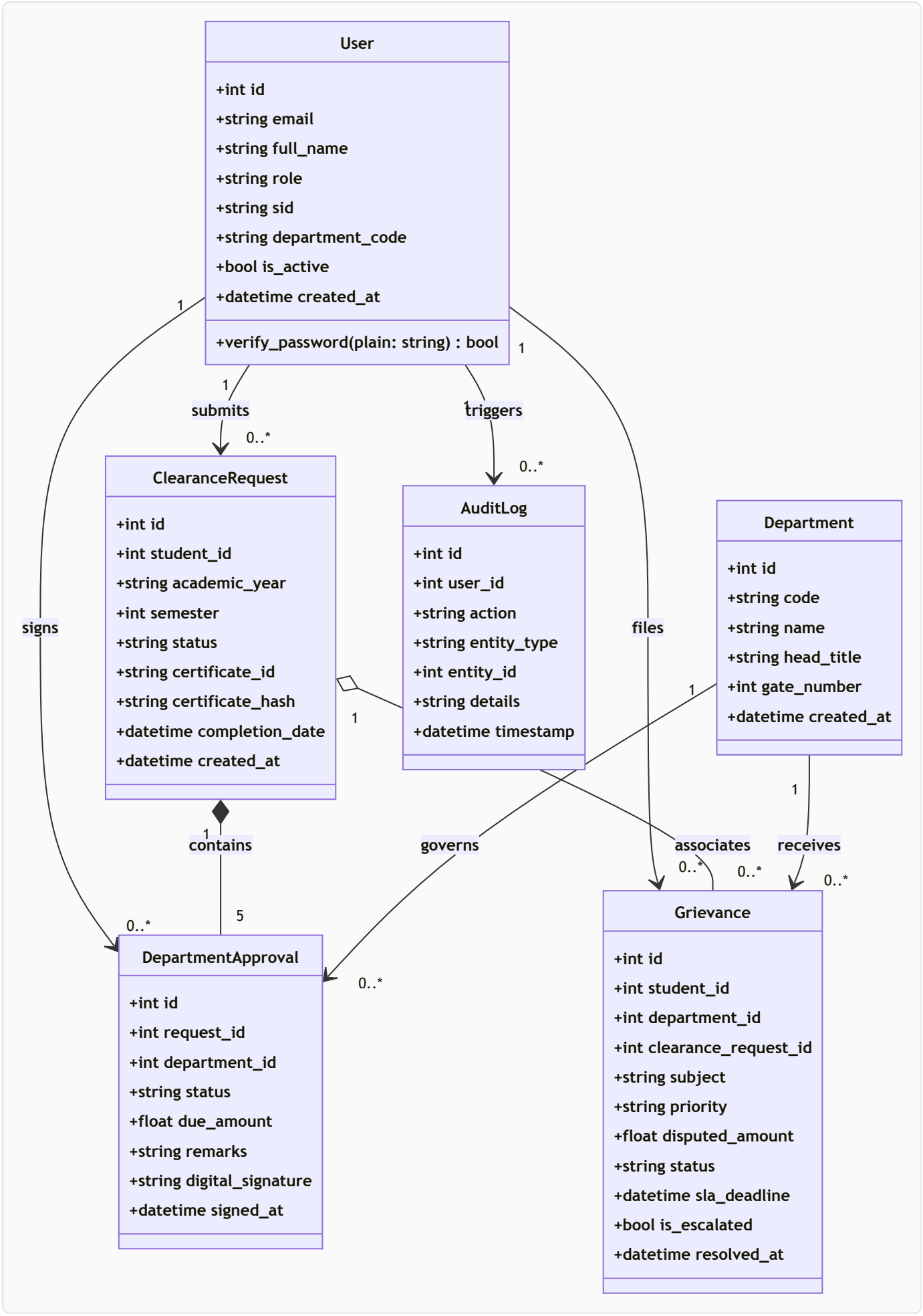


2. Static Structural Model: Class Diagram

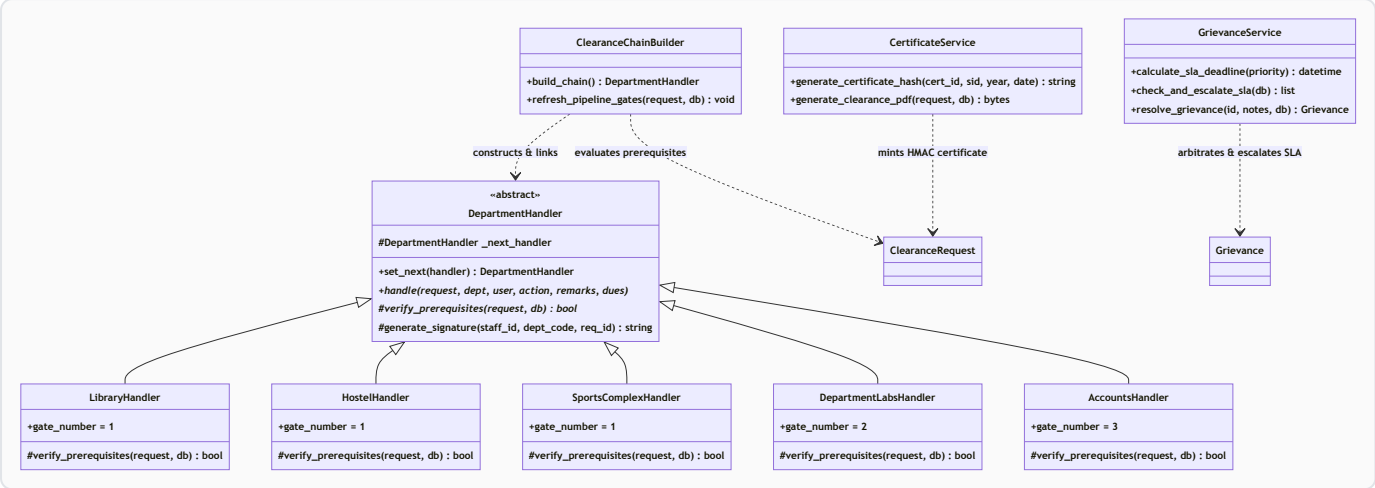
The static structural architecture of the system is divided into two focused, high-contrast class diagrams to ensure maximum visual clarity and readability:

1. **Core Domain Entity Model** (relational data models, attributes, methods, associations, and multiplicities)
2. **Behavioral Chain of Responsibility & Services Model** (handler hierarchy, gate verification, and service orchestration)

2.1 Core Domain Entity Class Diagram

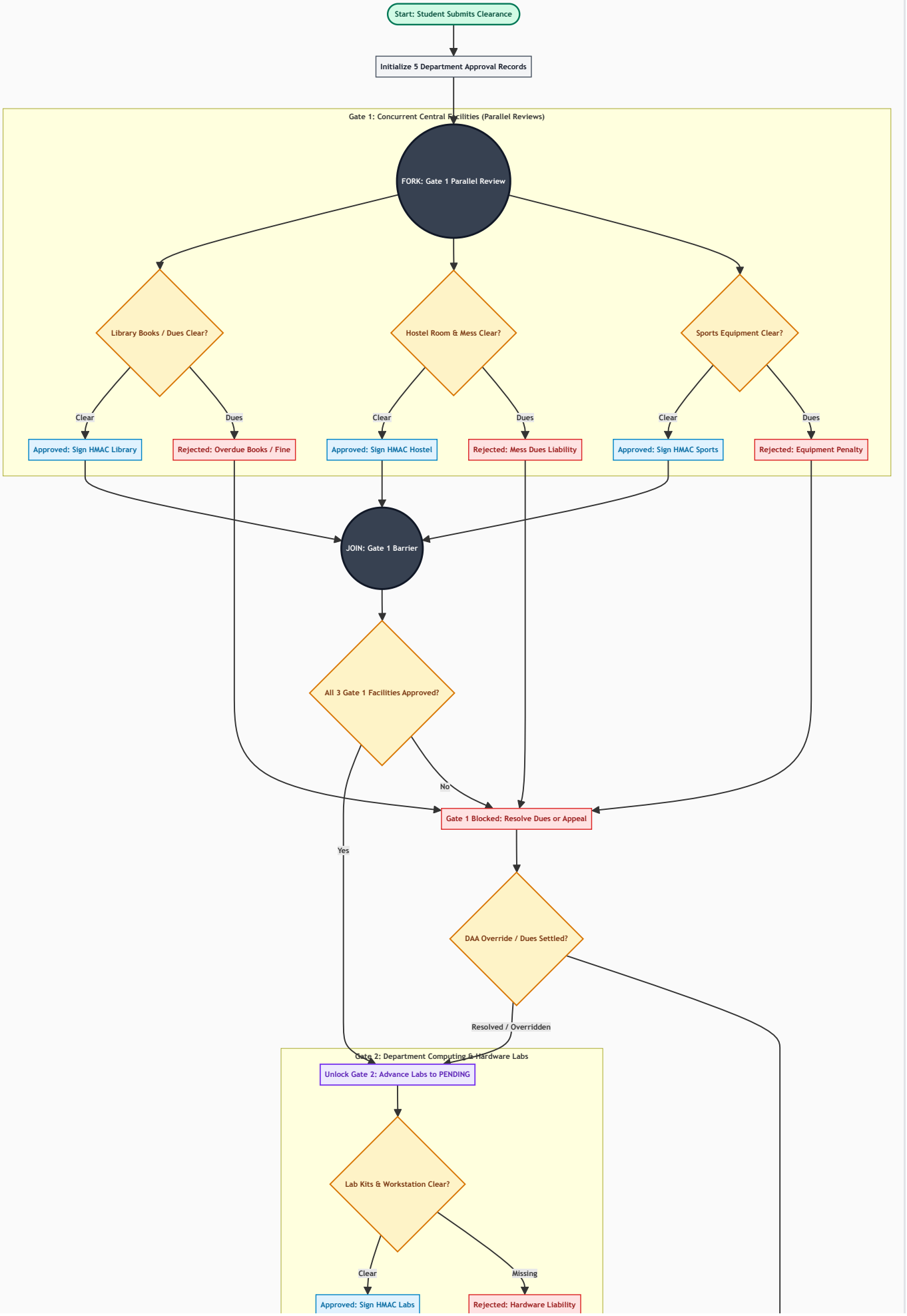

User
+int id
+string email
+string full_name
+string role
+string sid
+string department_code
+bool is_active
+datetime created_at
+verify_password(plain: string) : bool
submits
triggers
signs
files
ClearanceRequest
+int id
+int student_id
+string academic_year
+int semester
+string status
+string certificate_id
+string certificate_hash
+datetime completion_date
+datetime created_at
AuditLog
+int id
+int user_id
+string action
+string entity_type
+int entity_id
+string details
+datetime timestamp
Department
+int id
+string code
+string name
+string head_title
+int gate_number
+datetime created_at
contains
governs
associates
receives
DepartmentApproval
+int id
+int request_id
+int department_id
+string status
+float due_amount
+string remarks
+string digital_signature
+datetime signed_at
Grievance
+int id
+int student_id
+int department_id
+int clearance_request_id
+string subject
+string priority
+float disputed_amount
+string status
+datetime sla_deadline
+bool is_escalated
+datetime resolved_at

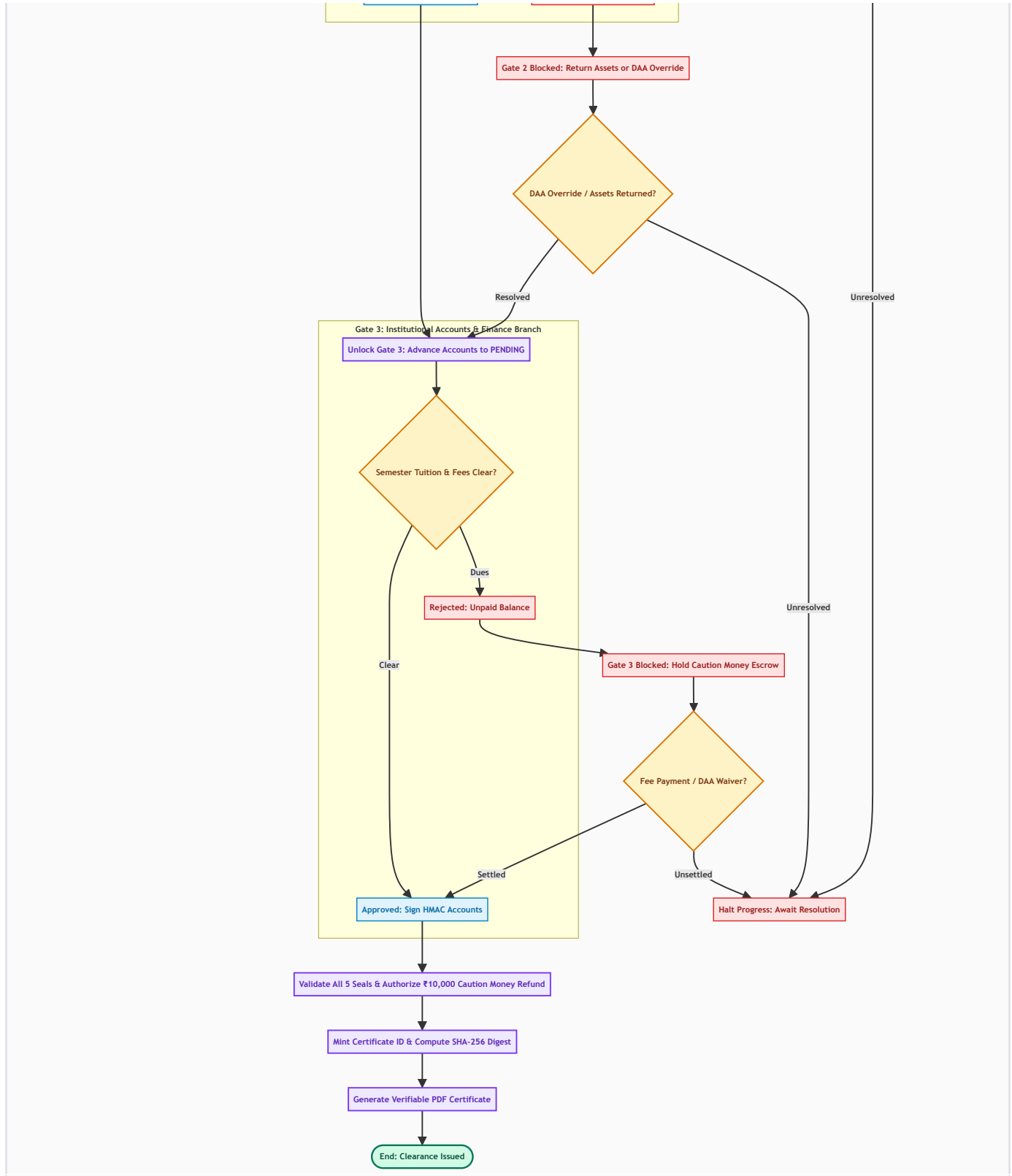
2.2 Behavioral Chain of Responsibility & Service Orchestration Diagram



3. Dynamic Behavioral Model: Activity Diagram

This activity diagram models the end-to-end dynamic workflow of a PEC clearance request, illustrating parallel concurrency forks, barrier joins, and sequential dependency gates:


Start: Student Submits Clearance
Initialize 5 Department Approval Records
Gate 1: Concurrent Central Facilities (Parallel Reviews)
FORK: Gate 1 Parallel Review
Library Books / Dues Clear?
Hostel Room & Mess Clear?
Sports Equipment Clear?
Clear
Dues
Clear
Dues
Clear
Dues
Approved: Sign HMAC Library
Rejected: Overdue Books / Fine
Approved: Sign HMAC Hostel
Rejected: Mess Dues Liability
Approved: Sign HMAC Sports
Rejected: Equipment Penalty
JOIN: Gate 1 Barrier
All 3 Gate 1 Facilities Approved?
No
Yes
Gate 1 Blocked: Resolve Dues or Appeal
DAA Override / Dues Settled?
Resolved / Overridden
Gate 2: Department Computing & Hardware Labs
Unlock Gate 2: Advance Labs to PENDING
Lab Kits & Workstation Clear?
Clear
Missing
Approved: Sign HMAC Labs
Rejected: Hardware Liability

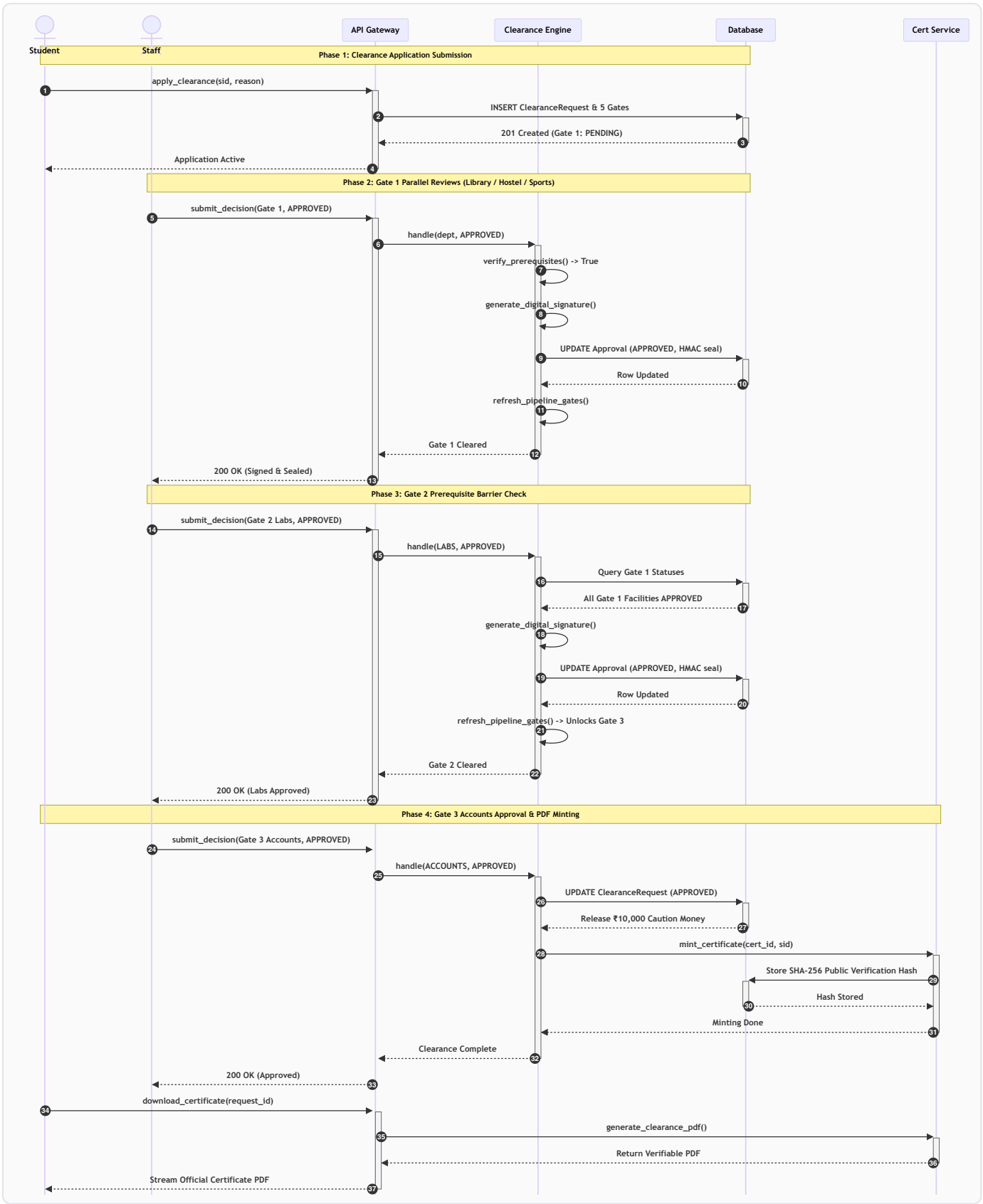

Gate 2 Blocked: Return Assets or DAA Override
DAA Override / Assets Returned?
Resolved
Unresolved
Gate 3: Institutional Accounts & Finance Branch
Unlock Gate 3: Advance Accounts to PENDING
Semester Tuition & Fees Clear?
Dues
Rejected: Unpaid Balance
Clear
Unresolved
Gate 3 Blocked: Hold Caution Money Escrow
Fee Payment / DAA Waiver?
Settled
Unsettled
Approved: Sign HMAC Accounts
Halt Progress: Await Resolution
Validate All 5 Seals & Authorize ₹10,000 Caution Money Refund
Mint Certificate ID & Compute SHA-256 Digest
Generate Verifiable PDF Certificate
End: Clearance Issued

4. Interaction Models: Sequence Diagrams

The dynamic interaction between external actors, API gateways, business logic engines, and database layers is presented across three focused sequence diagrams. Each diagram is structured with concise messages and clear phase notes for readability:

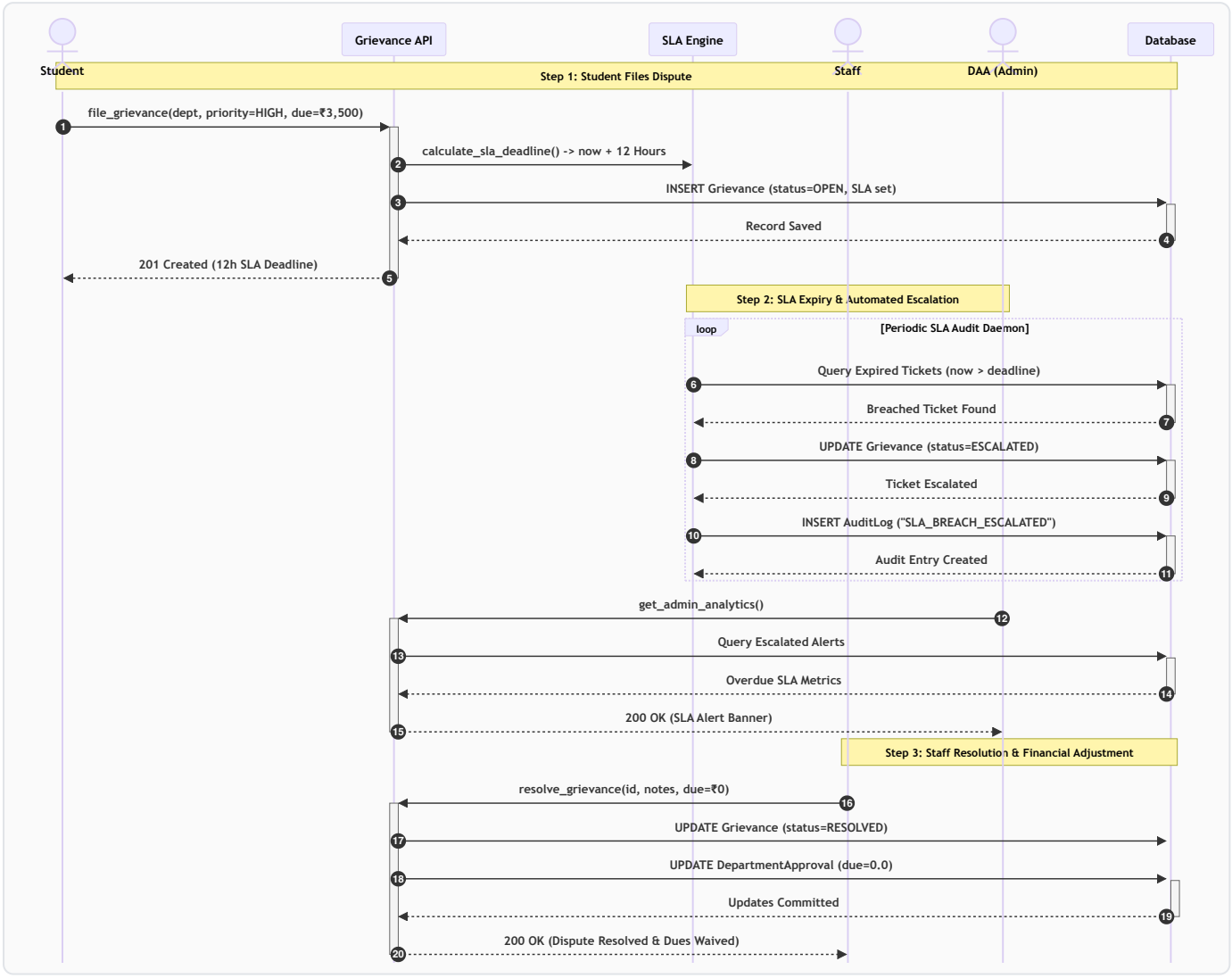
4.1 Sequence Diagram 1: Multi-Gate Approval & Cryptographic Minting

This diagram traces the standard clearance path from student initiation through Gate 1 parallel reviews, Gate 2 prerequisite verification, Gate 3 financial reconciliation, and final certificate minting:


Student
Staff
API Gateway
Clearance Engine
Database
Cert Service
Phase 1: Clearance Application Submission
apply_clearance(sid, reason)
INSERT ClearanceRequest & 5 Gates
201 Created (Gate 1: PENDING)
Application Active
Phase 2: Gate 1 Parallel Reviews (Library / Hostel / Sports)
submit_decision(Gate 1, APPROVED)
handle(dept, APPROVED)
verify_prerequisites() -> True
generate_digital_signature()
UPDATE Approval (APPROVED, HMAC seal)
Row Updated
refresh_pipeline_gates()
Gate 1 Cleared
200 OK (Signed & Sealed)
Phase 3: Gate 2 Prerequisite Barrier Check
submit_decision(Gate 2 Labs, APPROVED)
handle(LABS, APPROVED)
Query Gate 1 Statuses
All Gate 1 Facilities APPROVED
generate_digital_signature()
UPDATE Approval (APPROVED, HMAC seal)
Row Updated
refresh_pipeline_gates() -> Unlocks Gate 3
Gate 2 Cleared
200 OK (Labs Approved)
Phase 4: Gate 3 Accounts Approval & PDF Minting
submit_decision(Gate 3 Accounts, APPROVED)
handle(ACCOUNTS, APPROVED)
UPDATE ClearanceRequest (APPROVED)
Release ₹10,000 Caution Money
mint_certificate(cert_id, sid)
Store SHA-256 Public Verification Hash
Hash Stored
Minting Done
Clearance Complete
200 OK (Approved)
download_certificate(request_id)
generate_clearance_pdf()
Return Verifiable PDF
Stream Official Certificate PDF

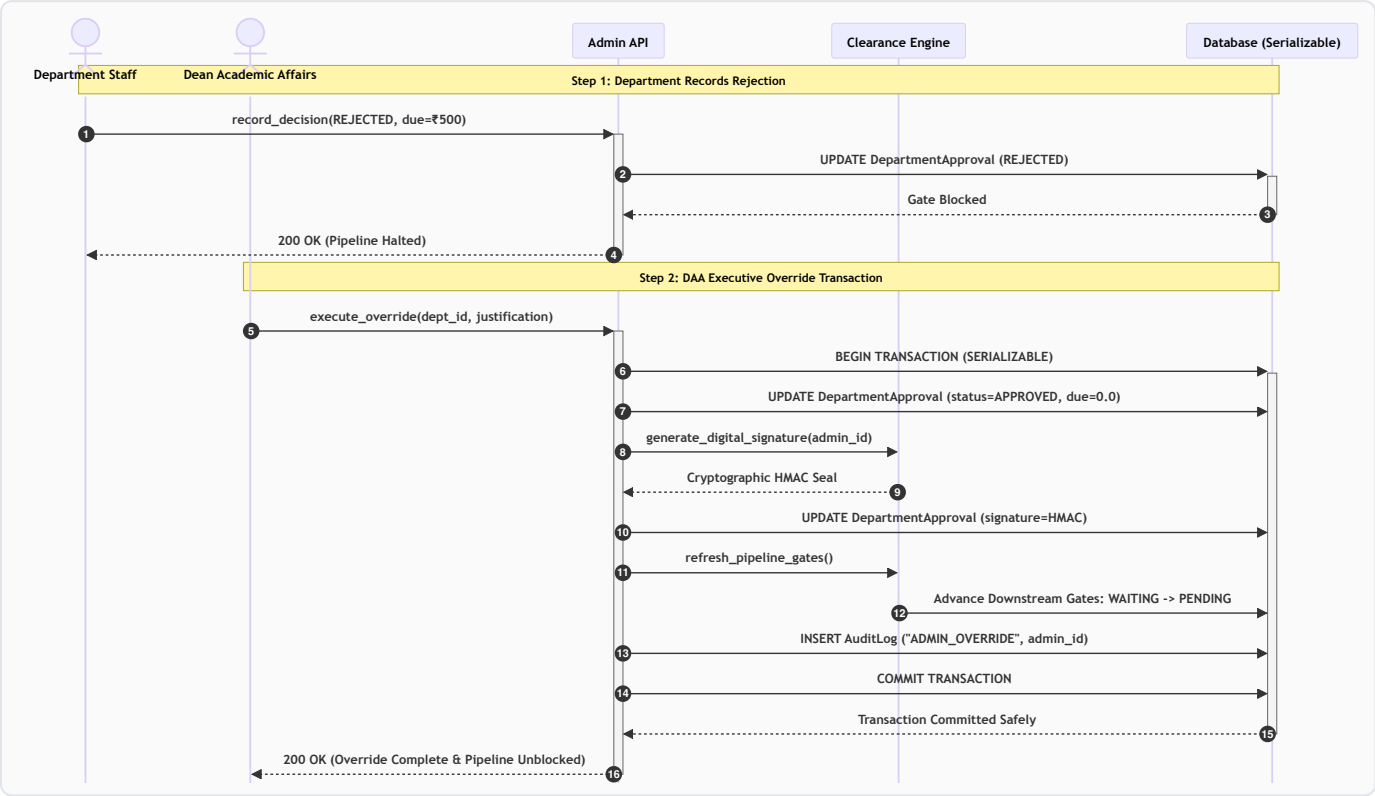
4.2 Sequence Diagram 2: Grievance Dispute Lifecycle & Automated SLA Escalation

This diagram models the grievance redressal lifecycle when a student disputes an assessed fine or liability, including dynamic SLA deadline assignment, background audit scanning, and DAA escalation:



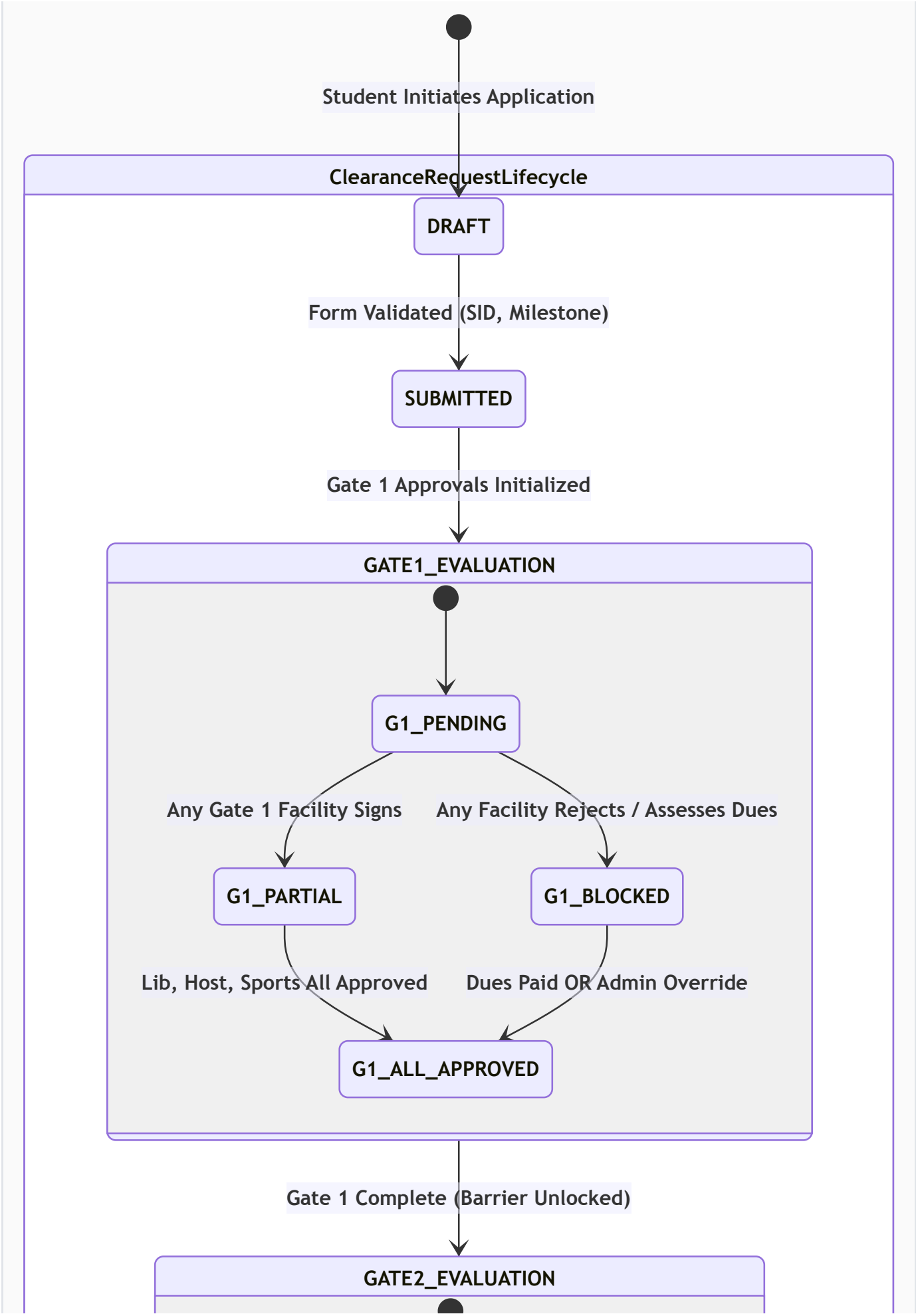
4.3 Sequence Diagram 3: Administrative Override & Concurrency Control

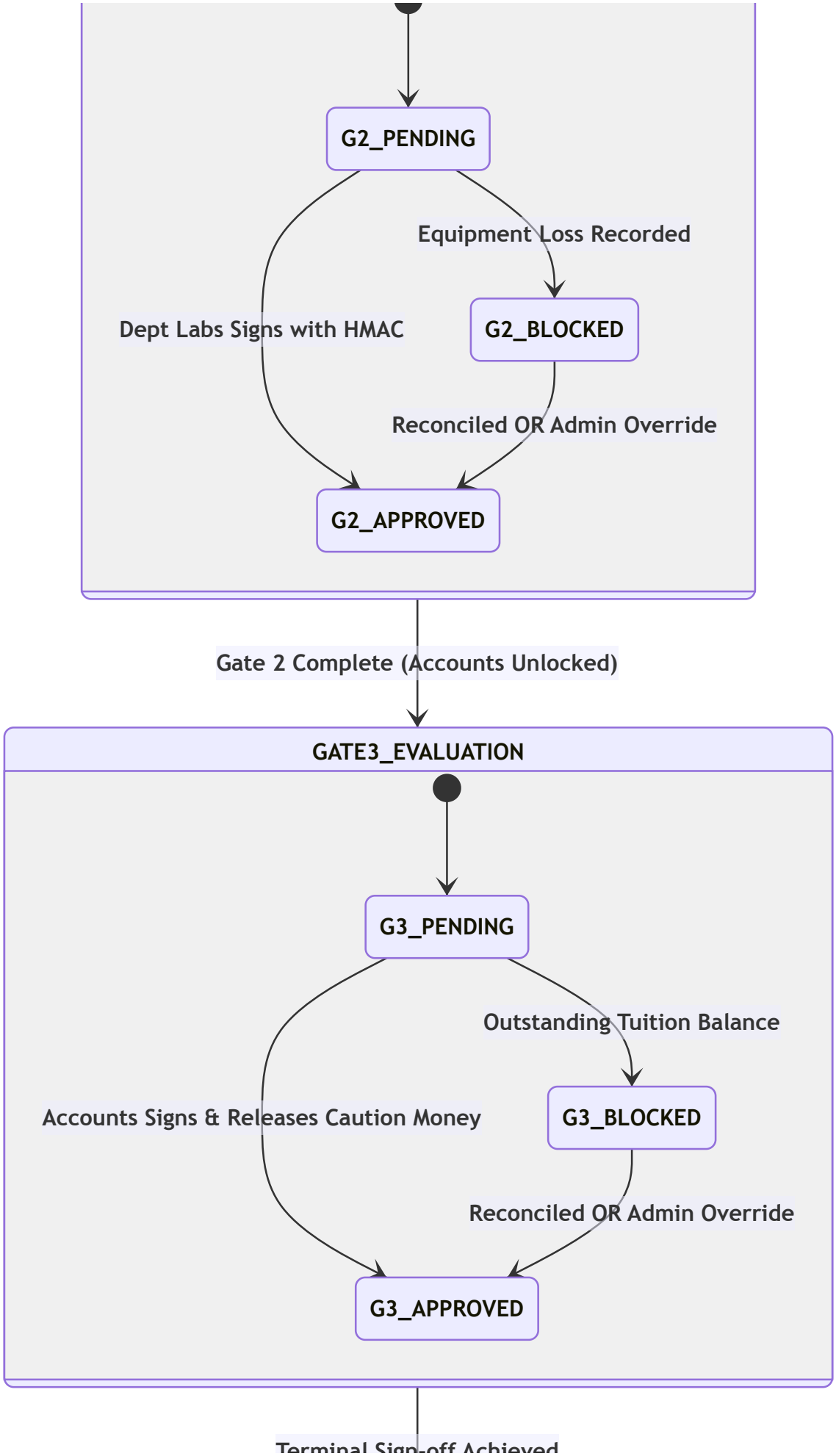
This diagram details the Dean Academic Affairs (DAA) executive override mechanism executed within a serializable database transaction to eliminate concurrency race conditions:

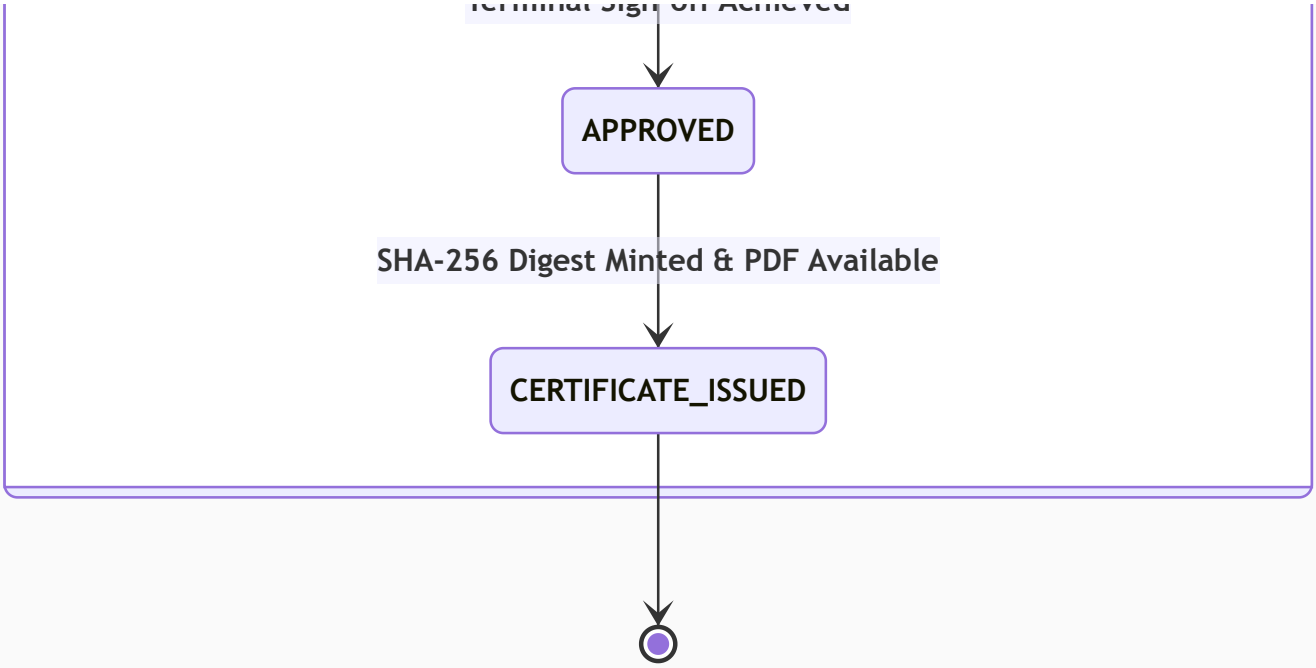


5. Behavioral State Machine Diagram

This state machine models the discrete status lifecycles of both the master clearance request and individual department approval gates:

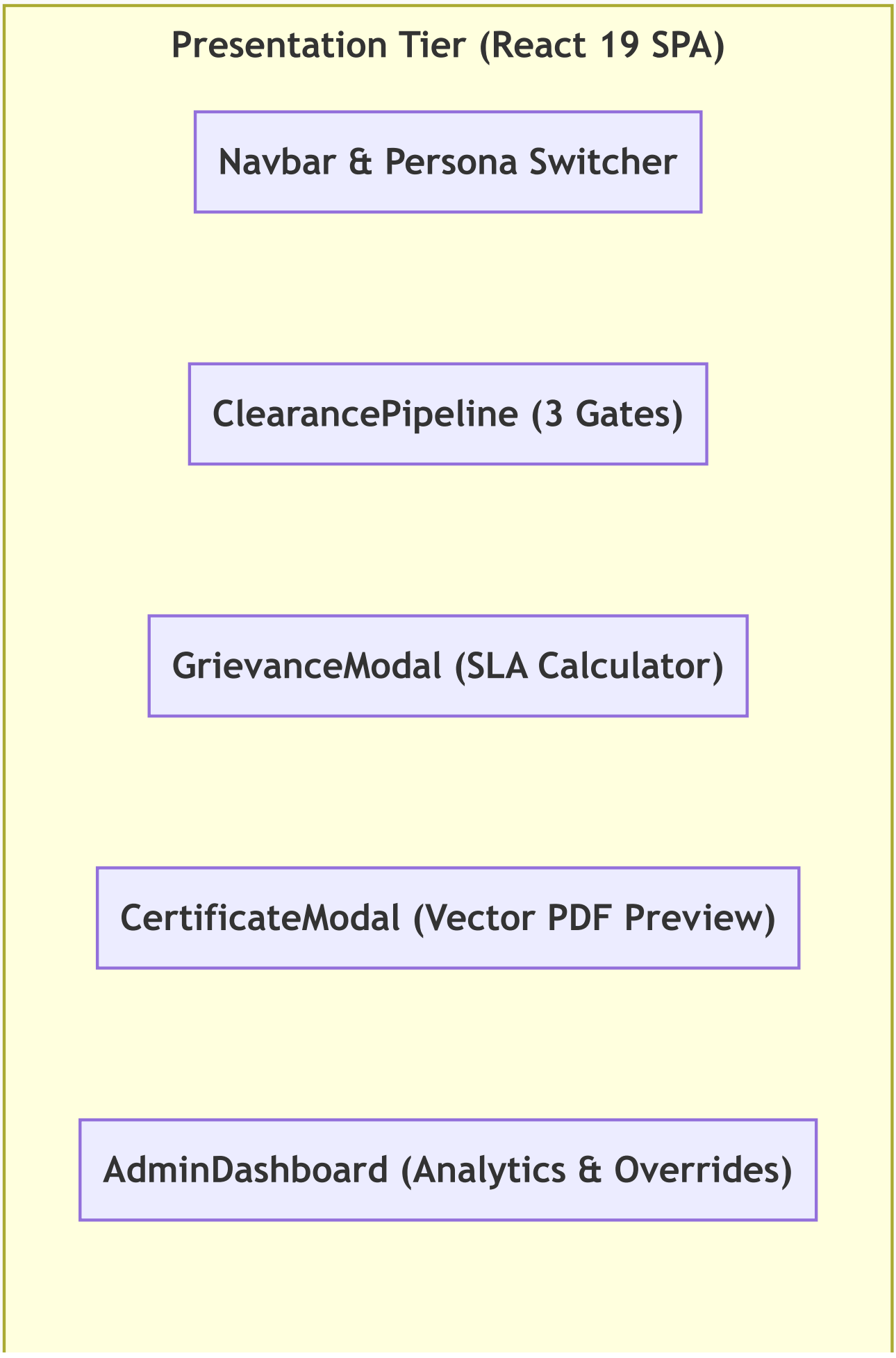

Student Initiates Application
ClearanceRequestLifecycle
DRAFT
Form Validated (SID, Milestone)
SUBMITTED
Gate 1 Approvals Initialized
GATE1_EVALUATION
G1_PENDING
Any Gate 1 Facility Signs
Any Facility Rejects / Assesses Dues
G1_PARTIAL
G1_BLOCKED
Lib, Host, Sports All Approved
Dues Paid OR Admin Override
G1_ALL_APPROVED
Gate 1 Complete (Barrier Unlocked)
GATE2_EVALUATION

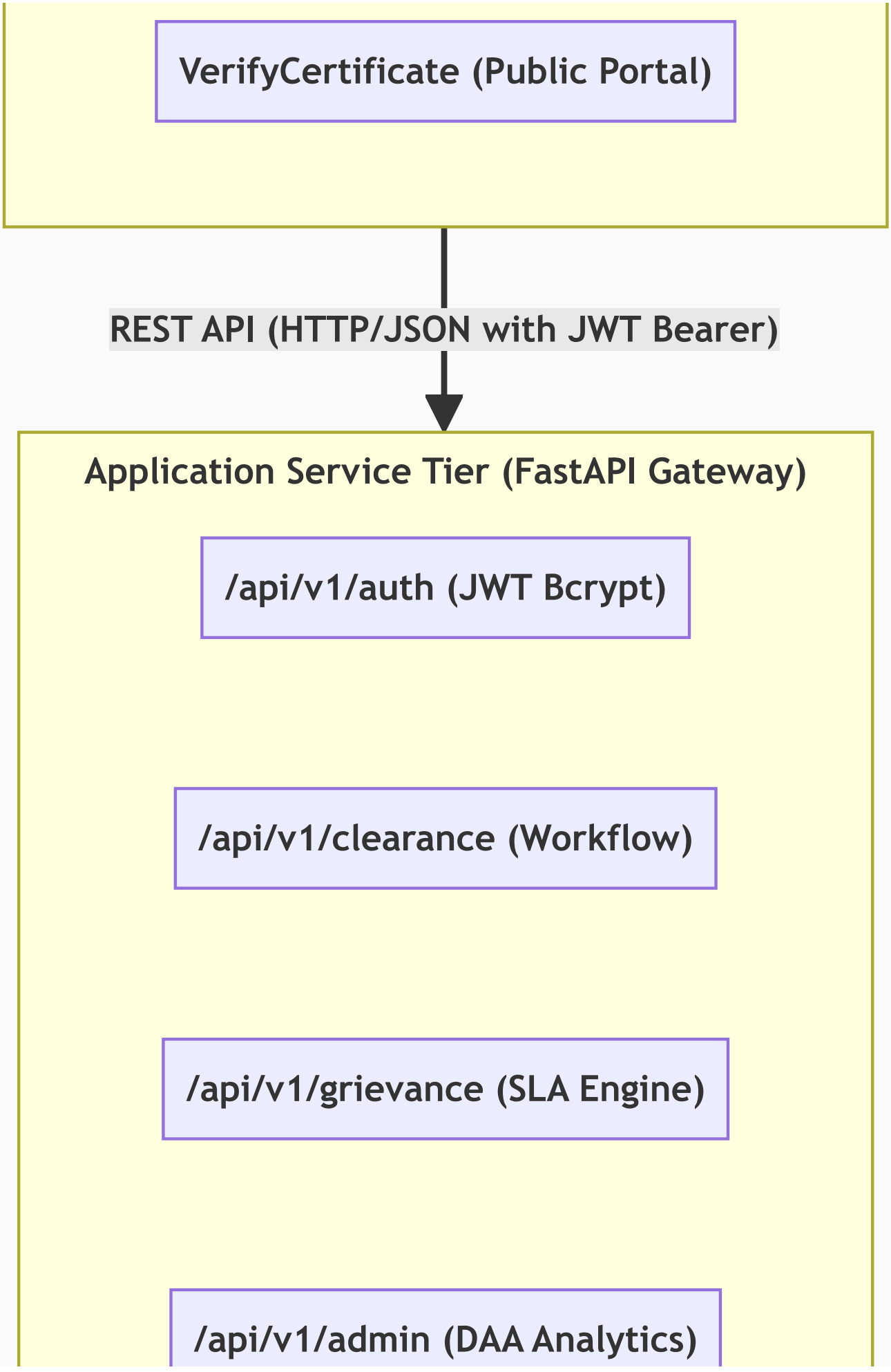

G2_PENDING
Equipment Loss Recorded
Dept Labs Signs with HMAC
G2_BLOCKED
Reconciled OR Admin Override
G2_APPROVED
Gate 2 Complete (Accounts Unlocked)
GATE3_EVALUATION
G3_PENDING
Outstanding Tuition Balance
Accounts Signs & Releases Caution Money
G3_BLOCKED
Reconciled OR Admin Override
G3_APPROVED
Terminal Sign-off Achieved

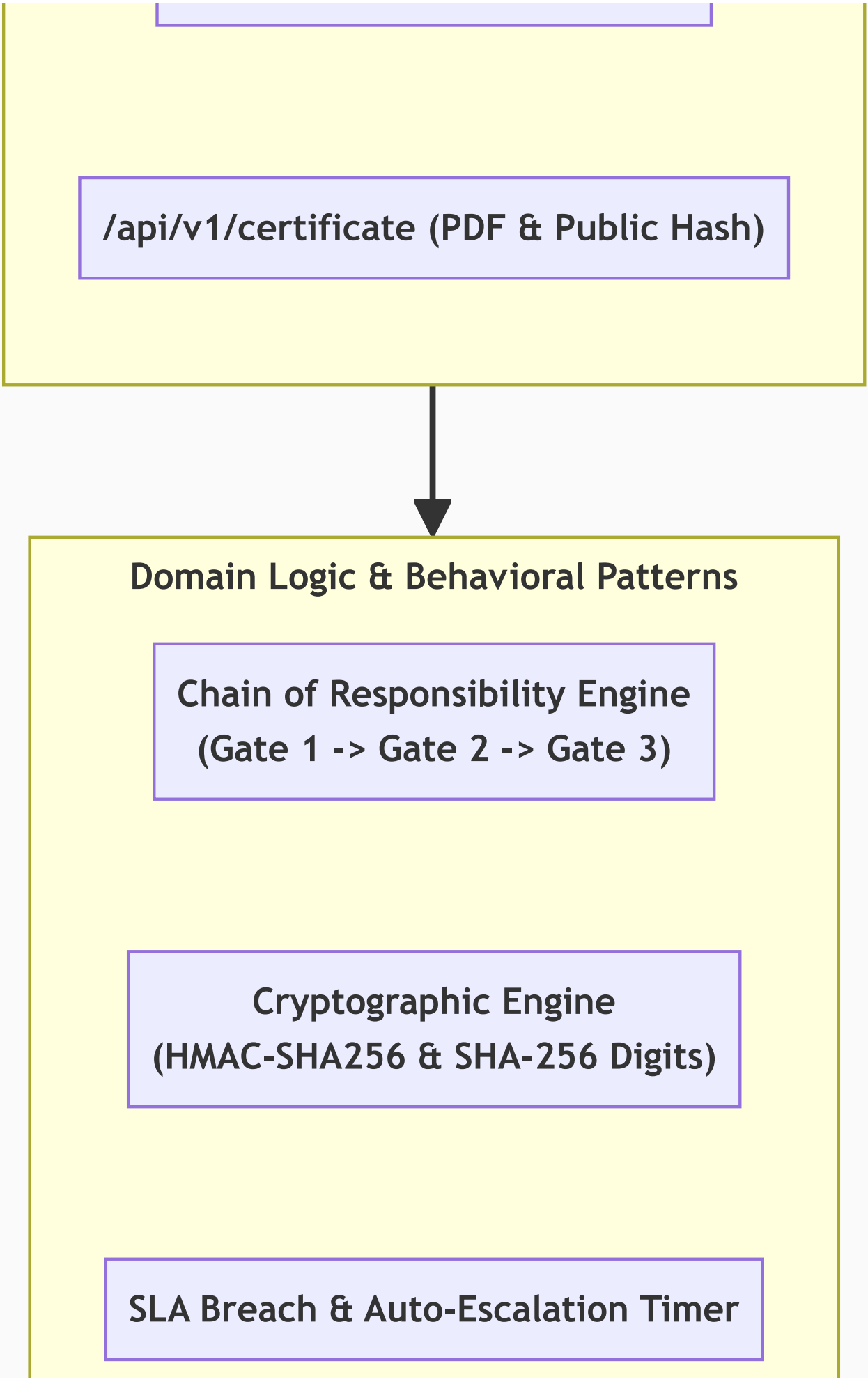

APPROVED
SHA-256 Digest Minted & PDF Available
CERTIFICATE_ISSUED

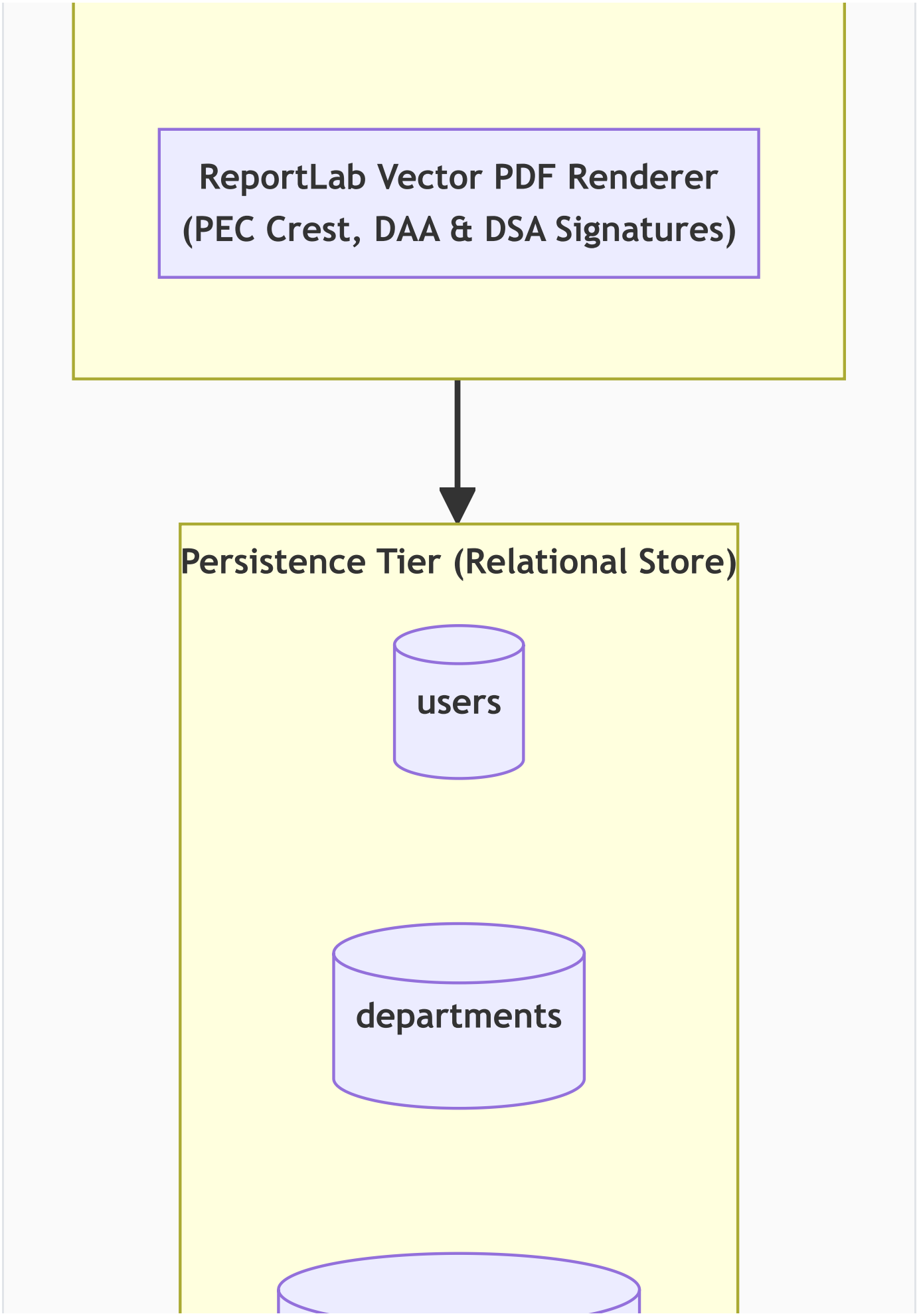
6. System Architecture & Component Deployment Diagram

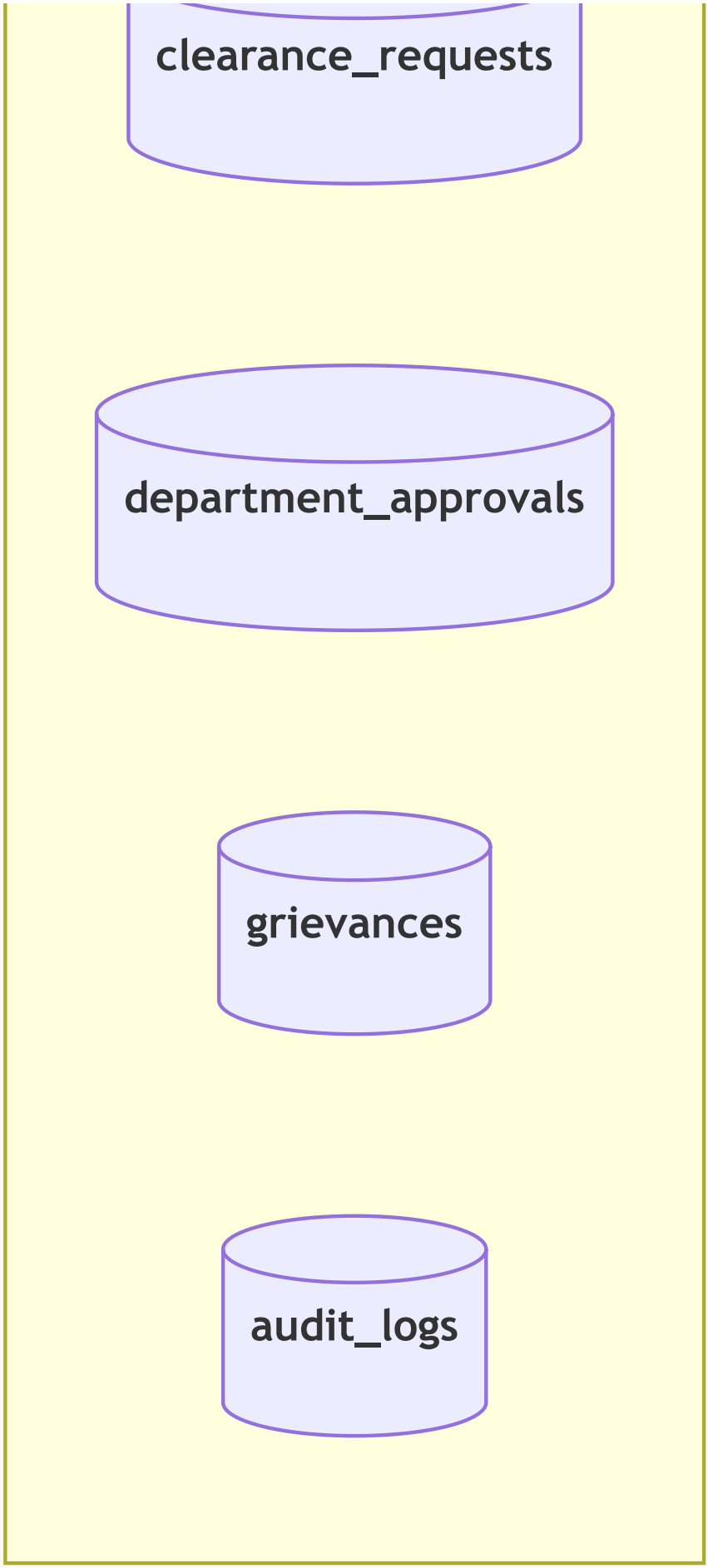
The component diagram defines the physical and logical boundaries of the system, verifying complete separation between presentation, application services, and persistent storage:


Presentation Tier (React 19 SPA)
Navbar & Persona Switcher
ClearancePipeline (3 Gates)
GrievanceModal (SLA Calculator)
CertificateModal (Vector PDF Preview)
AdminDashboard (Analytics & Overrides)


VerifyCertificate (Public Portal)
REST API (HTTP/JSON with JWT Bearer)
Application Service Tier (FastAPI Gateway)
/api/v1/auth (JWT Bcrypt)
/api/v1/clearance (Workflow)
/api/v1/grievance (SLA Engine)
/api/v1/admin (DAA Analytics)


/api/v1/certificate (PDF & Public Hash)
Domain Logic & Behavioral Patterns
Chain of Responsibility Engine
(Gate 1 -> Gate 2 -> Gate 3)
Cryptographic Engine
(HMAC-SHA256 & SHA-256 Digits)
SLA Breach & Auto-Escalation Timer


ReportLab Vector PDF Renderer
(PEC Crest, DAA & DSA Signatures)
Persistence Tier (Relational Store)
users
departments


clearance_requests
department_approvals
grievances
audit_logs